

Week 4: Environmental Hazards

JSS3 Intermediate Science — First Term Lesson Note (Deep-Dive Edition)

Introduction

This week studies environmental hazards — natural or human-influenced events that threaten people, property, and the environment — along with how communities can prepare for and reduce their impact.

Meaning and Types

Hazard	Description
Drought	A prolonged period of unusually low rainfall, leading to water shortages
Desertification	The process by which fertile land becomes desert, often due to drought, deforestation, or poor farming practices
Erosion	The wearing away of soil or rock by water, wind, or human activity
Flooding	The overflow of water onto normally dry land, often due to heavy rainfall or poor drainage

Causes and Prevention

Hazard	Common Causes	Prevention Measures
Drought	Low rainfall, climate change, overuse of water sources	Water conservation, irrigation planning, drought-resistant crops
Desertification	Deforestation, overgrazing, poor farming methods	Afforestation, controlled grazing, sustainable farming
Erosion	Deforestation, heavy rainfall, poor land use on slopes	Planting vegetation cover, terracing, building drainage channels
Flooding	Heavy rainfall, blocked drainage, poor urban planning	Clearing drainage channels, avoiding building on waterways, flood barriers

Disaster Preparedness

- Learn and follow early warning systems issued by relevant agencies
- Keep emergency supplies ready (water, food, first aid)
- Know evacuation routes and safe shelter locations
- Participate in community disaster drills where available

Community Safety Measures

Communities can reduce hazard risk through good drainage systems, tree-planting programmes, enforcement of building regulations away from flood-prone or erosion-prone areas, and public education campaigns about environmental hazards.

Real-Life Application

Northern Nigeria faces significant risk of desertification and drought, while erosion is a major concern in south-eastern states such as Anambra and Imo, and flooding regularly affects low-lying areas such as parts of Lagos and riverine communities during the rainy season.

Vocabulary

Word	Meaning
Desertification	The transformation of fertile land into desert-like conditions
Erosion	The gradual wearing away of soil or rock
Disaster preparedness	Actions taken in advance to reduce the impact of a disaster

Evaluation Questions

- 1 Define drought and desertification.
- 2 Explain the difference between erosion and flooding.
- 3 List two causes of desertification.
- 4 List two causes of flooding in urban areas.
- 5 Name two prevention measures for soil erosion.
- 6 What is disaster preparedness, and why does it matter?
- 7 Give one example of a Nigerian region affected by desertification.
- 8 Give one example of a Nigerian region affected by erosion.
- 9 Describe two community safety measures that can reduce flood risk.
- 10 In your own words, explain why understanding environmental hazards is important for community safety.

(Week 5 is Mid-Term Examination, and Week 6 is Mid-Term Break — no new content in these weeks.)